

Commercial Analysis and Strategic Insight

IBM Telco Customer Churn | Python and SQL

This report analyses customer churn in the IBM Telco dataset using SQL-based data preparation, commercial segmentation and an interpretable churn-risk model. The objective is to identify high-risk customer groups and translate model outputs into practical retention priorities.

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1 Executive summary

This project analyses the IBM Telco customer churn dataset to identify commercially meaningful churn patterns and build a simple customer risk-scoring framework. SQL was used to query, clean and segment the data, while Python was used for exploratory analysis, visualisation and logistic regression modelling.

The raw dataset contained 7,043 customer records. After removing 11 rows with blank **TotalCharges** values, the final analytical sample contained 7,032 customers. Of these, 1,869 customers churned, giving an overall churn rate of 26.58%.

The analysis shows that churn is not evenly distributed across the customer base. It is concentrated among month-to-month customers, customers in their first year, electronic check users and customers with high monthly charges. Month-to-month customers had a churn rate of 42.71%, compared with 11.28% for one-year contracts and 2.85% for two-year contracts. Customers in their first 12 months had a churn rate of 48.54%, while customers with more than 48 months of tenure had a churn rate of 9.64%.

A logistic regression model was then used to estimate churn probability. The model produced a ROC AUC of 0.8395, indicating useful ranking power for customer prioritisation. The predicted risk bands were commercially meaningful: the low-risk group had an actual churn rate of 4.54%, compared with 23.23% for the medium-risk group and 56.29% for the high-risk group.

The main recommendation is to prioritise retention activity towards early-tenure, month-to-month customers, especially where monthly charges are high or payment behaviour is associated with elevated churn risk. The output should be interpreted as churn-risk prioritisation rather than a live operational forecasting system.

Table 1: Executive summary metrics

Metric	Value
Customers analysed	7,032
Churned customers	1,869
Overall churn rate	26.58%
Model ROC AUC	0.8395
High-risk band churn rate	56.29%
Low-risk band churn rate	4.54%

2 Data and approach

The project uses the public IBM Telco customer churn dataset, which contains customer demographic, account, service and churn information. The objective is to use the dataset as a commercial analytics case study rather than to build a production churn system.

The analysis followed three stages:

- **SQL data preparation:** the raw CSV file was queried using DuckDB SQL. Customer fields

were cleaned, churn was converted into a binary flag, and commercial grouping variables were created.

- **Commercial segmentation:** churn rates were analysed across contract type, tenure band, payment method and monthly charge band to identify customer groups with elevated churn.
- **Churn-risk modelling:** a logistic regression model was used to estimate customer churn probability and allocate customers into low, medium and high-risk bands.

The dataset is static and does not contain transaction-level time series. Therefore, the model predicts churn probability using customer attributes, rather than forecasting churn dynamically through time.

Table 2: Data cleaning summary

Metric	Value
Raw customer records	7,043
Rows removed due to blank <code>TotalCharges</code>	11
Final analytical sample	7,032
Final churned customers	1,869
Final non-churned customers	5,163

3 Churn profile and commercial segmentation

3.1 Contract type

Contract type is the clearest churn differentiator in the dataset. Month-to-month customers had a churn rate of 42.71%, compared with 11.28% for one-year contracts and 2.85% for two-year contracts.

This suggests that contract structure is a central retention lever. Customers on rolling monthly contracts have more flexibility to leave, while longer-term contracts appear associated with stronger retention.

Table 3: Churn by contract type

Contract type	Customers	Churned	Churn rate	Avg monthly charge
Month-to-month	3,875	1,655	42.71%	66.40
One year	1,472	166	11.28%	65.08
Two year	1,685	48	2.85%	60.87

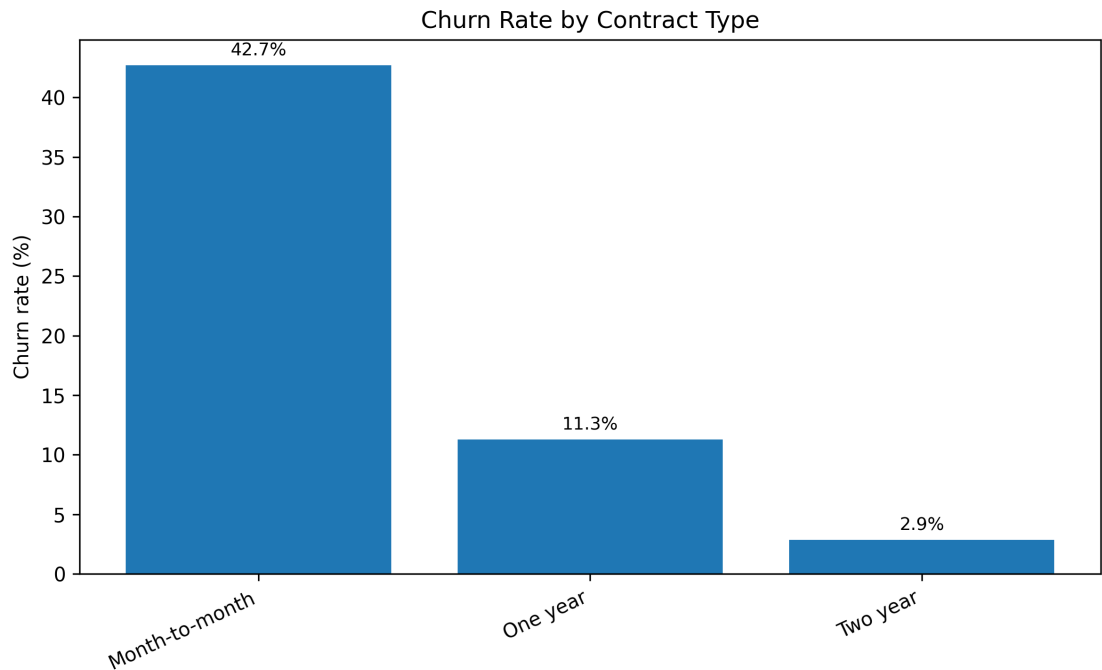


Figure 1: Churn rate by contract type. Month-to-month customers show a materially higher churn rate than customers on one-year or two-year contracts.

3.2 Tenure band

Churn is also strongly related to customer tenure. Customers in their first 12 months had a churn rate of 48.54%, while customers with more than 48 months of tenure had a churn rate of 9.64%.

This indicates that churn risk is highest early in the customer lifecycle. Retention actions should therefore focus on onboarding, early service experience and first-year engagement, rather than only targeting long-tenure customers after dissatisfaction has accumulated.

Table 4: Churn by tenure band

Tenure band	Customers	Churned	Churn rate	Avg monthly charge
0–12 months	2,058	999	48.54%	56.13
12–24 months	1,047	309	29.51%	60.85
24–48 months	1,624	339	20.87%	65.73
48+ months	2,303	222	9.64%	73.68

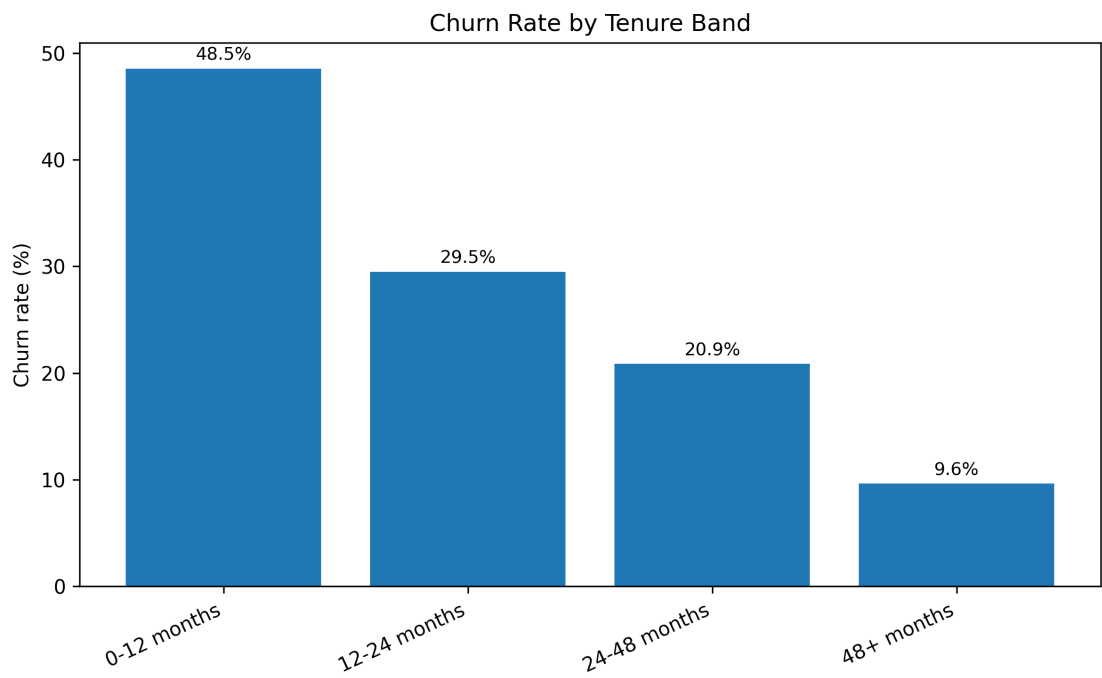


Figure 2: Churn rate by tenure band. Churn is concentrated in the first year and declines materially as tenure increases.

3.3 Payment method

Payment method is another useful churn indicator. Electronic check customers had a churn rate of 45.29%, compared with 16.73% for automatic bank transfer customers and 15.25% for automatic credit card customers.

This should not be interpreted as proof that payment method causes churn. However, it is a useful commercial signal. Electronic check may reflect customer profile differences, payment friction or weaker account engagement. It is therefore a sensible area for customer journey review.

Table 5: Churn by payment method

Payment method	Customers	Churned	Churn rate	Avg monthly charge
Electronic check	2,365	1,071	45.29%	76.26
Mailed check	1,604	308	19.20%	43.96
Bank transfer (automatic)	1,542	258	16.73%	67.21
Credit card (automatic)	1,521	232	15.25%	66.52

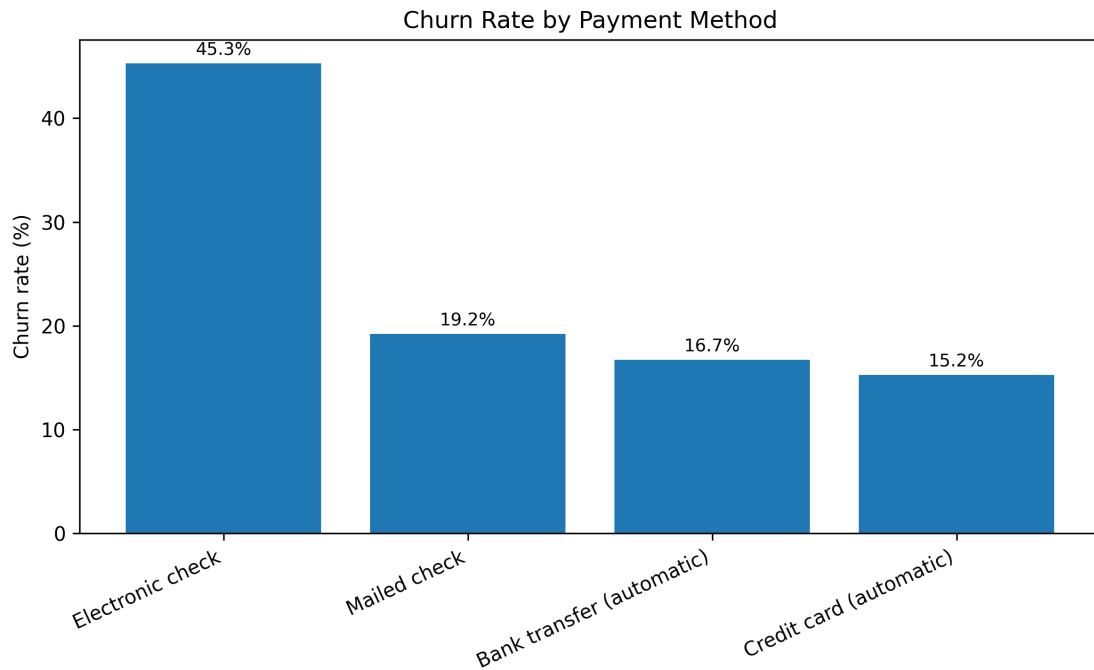


Figure 3: Churn rate by payment method. Electronic check users show substantially higher churn than customers using automatic payment methods.

3.4 Monthly charge band

Monthly charge band provides an additional commercial view. High monthly charge customers had a churn rate of 35.50%, compared with 23.69% for medium-charge customers and 10.90% for low-charge customers.

This suggests that price, perceived value or product complexity may be relevant to customer retention. High-charge customers may generate more revenue, but they also appear to require more focused retention management.

Table 6: Churn by monthly charge band

Monthly charge band	Customers	Churned	Churn rate	Avg monthly charge
High	3,589	1,274	35.50%	90.19
Medium	1,718	407	23.69%	54.72
Low	1,725	188	10.90%	22.00

4 High-risk customer segments

The strongest commercial insights come from combining multiple risk indicators. The highest-risk groups are largely defined by month-to-month contracts, short tenure and high monthly charges.

The most actionable high-risk segment is month-to-month customers in their first 12 months with high monthly charges and electronic check payment. This segment contains 562 customers and has a churn rate of 71.00%. This is a stronger commercial target than very small high-churn segments

because it combines both high risk and meaningful customer volume.

Table 7: Selected high-risk customer segments

Contract type	Tenure band	Charge band	Payment method	Customers	Churned	Churn rate
Month-to-month	0–12 months	High	Bank transfer (automatic)	73	58	79.45%
Month-to-month	0–12 months	High	Electronic check	562	399	71.00%
Month-to-month	0–12 months	High	Credit card (automatic)	71	46	64.79%
Month-to-month	0–12 months	High	Mailed check	112	67	59.82%
Month-to-month	12–24 months	High	Electronic check	259	150	57.92%

The segmentation results show that churn risk compounds when multiple factors are present. A customer with short tenure, a month-to-month contract and high monthly charges is materially more exposed than the average customer. This provides a practical basis for prioritising retention campaigns.

5 Churn-risk model

A logistic regression model was trained to estimate churn probability. The model used numerical features such as tenure, monthly charges and total charges, alongside one-hot encoded categorical features such as contract type, payment method and service usage.

The model was intentionally kept simple. For a commercial analysis project, interpretability is more valuable than marginal predictive gains from a more complex model. The aim is to create a usable risk-ranking tool and support business recommendations, rather than to optimise a production machine learning system.

Table 8: Model validation metrics

Metric	Value
Accuracy	0.7355
Precision	0.5014
Recall	0.7923
ROC AUC	0.8395

The ROC AUC of 0.8395 indicates useful ranking performance. Recall is relatively high at 0.7923, which is appropriate for a retention use case where the business may prefer to identify a larger share of potential churners, even at the cost of contacting some customers who do not churn.

Precision is lower at 0.5014, which means not all customers flagged as churn risks would actually churn. This is not unusual in churn management, where the purpose of the model is often to prioritise attention rather than make a final decision automatically.

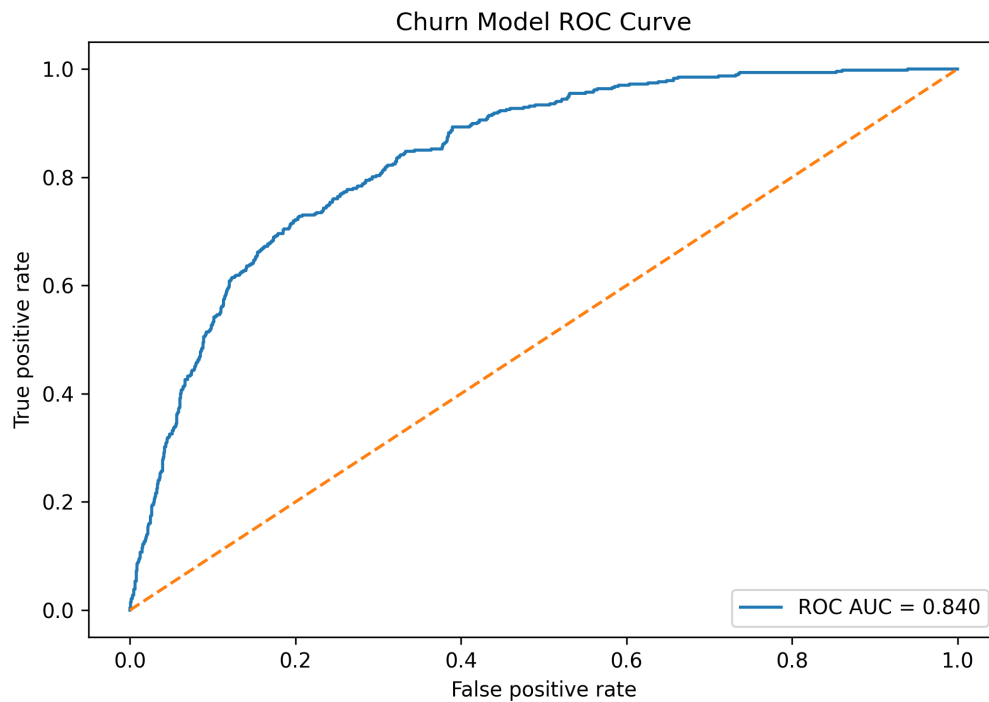


Figure 4: Churn model ROC curve. The ROC AUC of approximately 0.84 indicates that the model has useful ability to rank customers by churn risk.

6 Risk banding and retention prioritisation

The predicted churn probabilities were converted into three risk bands: low, medium and high. This makes the model output easier to use in a commercial setting. Instead of giving every customer a raw probability, the business can use risk bands to allocate retention activity.

The risk bands show clear separation in observed churn rates. The low-risk group had an actual churn rate of 4.54%, the medium-risk group had an actual churn rate of 23.23%, and the high-risk group had an actual churn rate of 56.29%.

Table 9: Customer risk band summary

Risk band	Customers	Churned	Churn rate	Avg monthly charge	Avg predicted churn probability
Low	2,973	135	4.54%	55.27	10.98%
Medium	1,666	387	23.23%	63.33	44.56%
High	2,393	1,347	56.29%	77.66	77.40%

The high-risk group has an actual churn rate more than twelve times higher than the low-risk group. This is the strongest practical output of the project: the model converts customer attributes into a usable retention prioritisation framework.

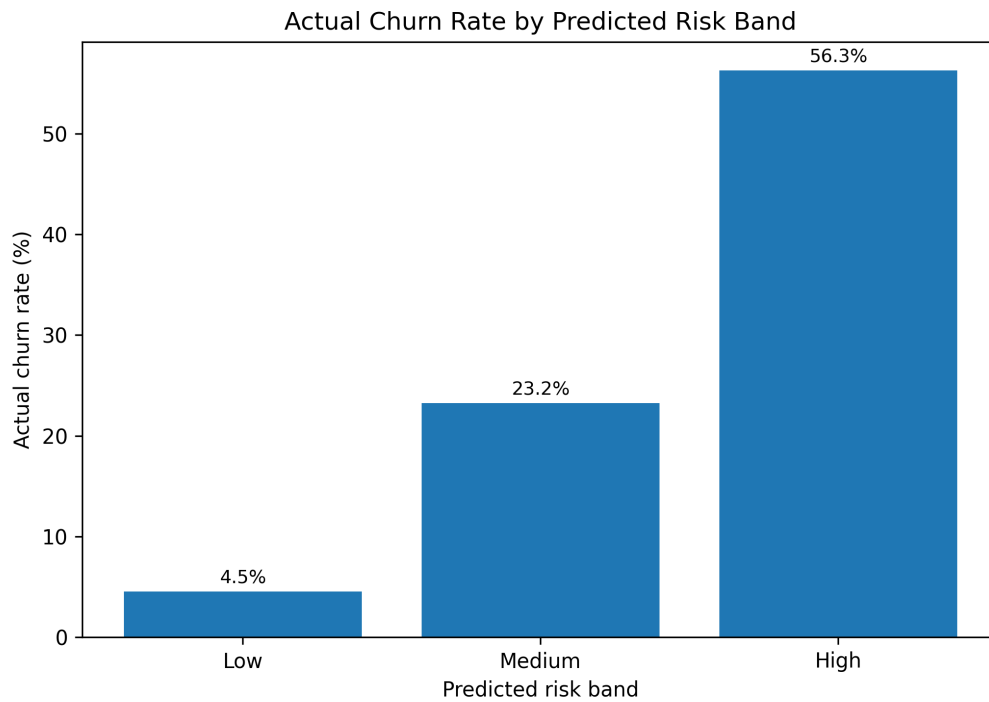


Figure 5: Actual churn rate by predicted risk band. The model separates customers into commercially meaningful risk groups.

7 Model interpretation

The chart below shows the largest positive logistic regression coefficients. These should be interpreted as model associations rather than causal effects. Several customer attributes are correlated, so the coefficient view is used for directional interpretation only.

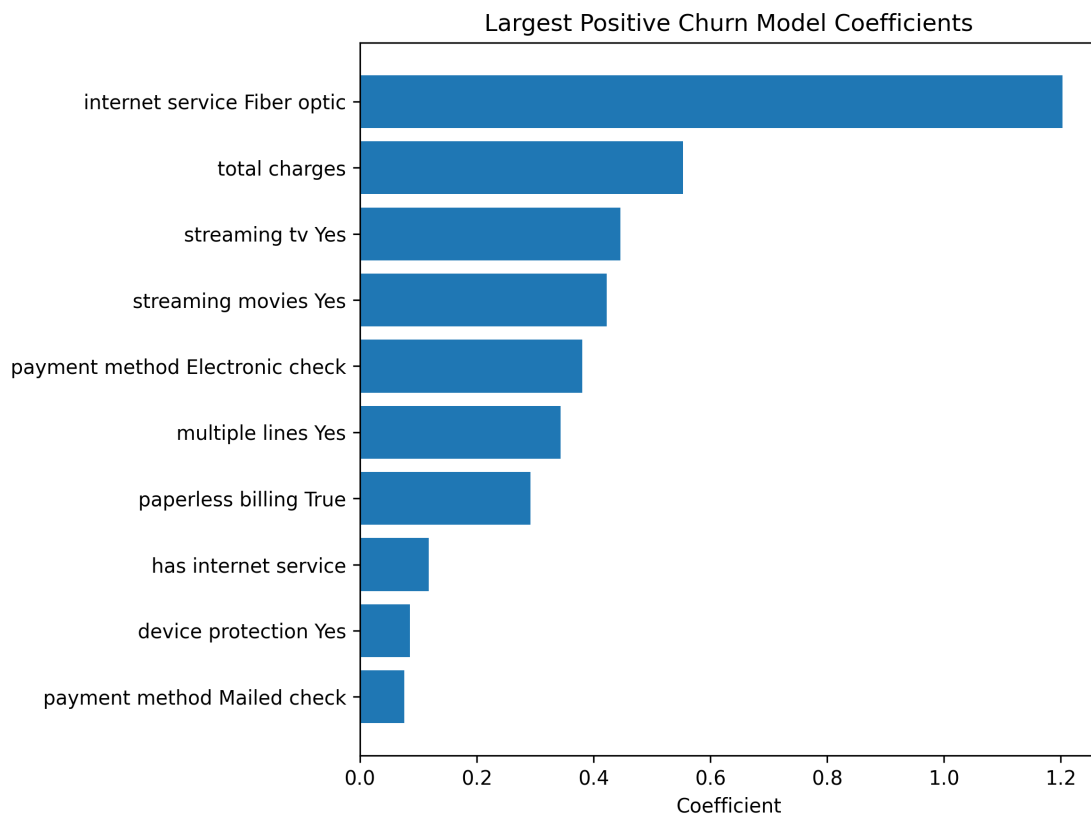


Figure 6: Largest positive churn model coefficients. Fiber optic service, higher total charges, streaming services and electronic check payment are associated with higher predicted churn in the model.

The coefficient view broadly supports the segmentation analysis by highlighting service mix, payment method and customer value variables as important churn-risk indicators. However, the segmentation results remain more directly actionable because they translate customer characteristics into clear churn rates and customer counts.

8 Commercial recommendations

The analysis supports four main commercial recommendations.

8.1 Prioritise first-year customers

Customers in their first 12 months have the highest tenure-related churn rate at 48.54%. Retention activity should therefore start early in the customer lifecycle. Onboarding, service activation, customer support quality and first-bill experience are likely to be important operational areas.

8.2 Target month-to-month customers

Month-to-month customers have a churn rate of 42.71%, materially above the overall churn rate of 26.58%. The business should test targeted offers that encourage high-risk monthly customers to

move towards longer contracts, particularly where the customer also has high monthly charges.

8.3 Review payment and billing friction

Electronic check customers have a churn rate of 45.29%, significantly higher than automatic payment users. The business should investigate whether this reflects payment friction, billing issues or customer profile differences. Nudges towards automatic payment methods may be worth testing, but this should be treated as a hypothesis rather than an assumed causal solution.

8.4 Use risk bands for retention prioritisation

The model-generated risk bands provide a practical targeting tool. Rather than applying the same retention activity across the full customer base, the business can prioritise high-risk customers and allocate lighter-touch engagement to medium-risk customers. This should improve efficiency by focusing retention spend where churn risk is most concentrated.

9 Limitations and further work

The analysis has several limitations.

First, the dataset is static. The model predicts churn probability from customer attributes, but it does not forecast churn dynamically over time. A live churn system would require dated customer interactions, billing events, service usage and retention campaign outcomes.

Second, the dataset is public and simulated, so the findings should be interpreted as a demonstration of analytical workflow rather than evidence about a specific live telecoms business.

Third, the analysis identifies associations rather than causal effects. For example, electronic check is associated with higher churn, but this does not prove that payment method causes churn. Customers who use electronic check may differ in other ways that are not fully captured in the dataset.

Further work could include time-based customer behaviour, uplift modelling for retention campaigns, cost-benefit analysis of offers and customer lifetime value integration. This would allow the business to prioritise customers not only by churn probability, but also by expected commercial value.

10 Conclusion

This project shows how a simple SQL and Python workflow can turn a customer churn dataset into commercial insight. The analysis identifies clear churn concentrations across contract type, tenure, payment method and monthly charge bands. It also shows that an interpretable logistic regression model can separate customers into useful risk bands.

The main takeaway is that churn risk is commercially concentrated rather than evenly distributed. The most actionable target groups are early-tenure, month-to-month customers with high monthly charges, particularly where payment behaviour is associated with elevated churn risk. These findings provide a practical basis for retention prioritisation, customer journey review and targeted commercial intervention.

References

- [1] IBM Sample Data Sets. *Telco Customer Churn Dataset*. Public customer churn dataset used for analytics and modelling demonstration.